「Morera Thm,

「Thm. Let $f: G \to \mathbb{C}$ be a continuous function on a region G such that

$$\int_T f = 0$$

for every triangular path T in G, then f is analytic in G.

Proof. Enough to show that f has a primitive. WLOG, let $G = B_R(a)$. Define

$$F(z) = \int_{[a,z]} f(w)\,dw, \quad \text{where } [a,z] : (1-t)a + tz, \quad 0 \le t \le 1.$$

Let $z_0 \in G$ be fixed. Then, $F(z) = \int_{[a,z_0]} f + \int_{[z_0,z]} f$, so

$$\frac{F(z) - F(z_0)}{z - z_0} = \frac{1}{z - z_0} \cdot \int_{[z_0,z]} f$$

Therefore, $$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0} \cdot \int_{[z_0,z]} f - f(z_0)$$

$$= \frac{1}{z - z_0} \cdot \int_{[z_0,z]} [f(w) - f(z_0)]\,dw.$$

Now, $$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| = \left| \frac{1}{z - z_0} \cdot \int_{[z_0,z]} [f(w) - f(z_0)]\,dw \right| \le |f(z) - f(z_0)| \to 0,$$